# Quantum Anomalies with Lara Anderson

**Podcast:** Startalk Podcast

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# AI Executive Summary

1. Core Thesis This episode of Star Talk explores the intersection of confirmed particle physics (Higgs boson) and unverified theoretical frameworks (String Theory) through a Q&A with Professor Laura Anderson. While the Higgs boson has been experimentally verified as the mechanism for mass generation within the Standard Model, String Theory remains a mathematically consistent but experimentally elusive candidate for Quantum Gravity. The dialogue highlights the difficulty of visualizing higher dimensions, the potential for gravity to be diluted across extra spatial dimensions (resolving the hierarchy problem), and the distinction between mathematical constructs and physical reality in modern physics.

2. Key Arguments & Data

3. **Higgs Boson Status:** Considered "thoroughly discovered" with Nobel Prize validation; properties aligning perfectly with Standard Model predictions; no current doubts regarding its role in mass generation.
4. **Experimental Resolution:** The Large Hadron Collider (LHC) probes energy scales up to 13 TeV, corresponding to distance scales of approximately $10^{-19}$ meters ($1 \times 10^{-19}$ m). To probe smaller structures requires higher energy/shorter wavelengths (analogous to x-ray diffraction vs. visible light).
5. **String Theory Geometry:** Strings are 1D objects, but movement through time creates a "world sheet" (2D surface). The theory is mathematically consistent regarding quantum gravity only when it accounts for exactly 10 dimensions ($D=10$) to cancel out quantum anomalies.
6. **Hierarchy Problem & Extra Dimensions:** Gravity is many orders of magnitude weaker than electromagnetic forces. A leading proposal suggests gravity's apparent weakness is due to it "leaking" or diluting into large extra dimensions, potentially explaining why other forces remain localized in our 3D space-time.
7. **Visualization of Higher Dimensions:** 4D objects can be conceptualized via their 3D shadows (like Tesseracts). A 2D being

cannot perceive the 3rd dimension unless the extra dimension is macroscopically "large" enough to intersect with; if curled up at string scales, they are invisible to current resolution.

8. **Discrete vs. Smooth Space-Time:** Alternatives like Loop Quantum Gravity (LQG) attempt to discretize space-time into building blocks to solve singularities. This contrasts with the smooth geometry assumed in many String Theory calculations.

9. **Time & Measurement:** Time is fundamentally a tool for sequencing events; it cannot be meaningfully measured in a static, non-repeating universe. There is no universal clock; time flows differently across gravitational fields per Special Relativity.

10. Actionable Takeaways

11. **Follow High-Energy Research:** Monitor experiments at the LHC and future colliders for "missing energy" signatures or anomalies that could indicate gravity leaking into extra dimensions, as these offer a potential path to test String Theory within the next few decades.
12. **Review Theoretical Texts:** Familiarize yourself with mathematical proofs of anomaly cancellation (D=10) to understand why 10-dimensional string theory is mathematically rigid and preferred over simpler models.
13. **Visualize Space-Time Concepts:** Utilize shadow analogies (e.g., 3D objects casting shadows on 2D planes) or Tesseract visualizations to better grasp how higher spatial dimensions might function without being directly perceivable.
14. **Evaluate String Theory Skepticism:** Acknowledge that String Theory is a powerful mathematical framework for unifying quantum mechanics and general relativity, but it currently lacks empirical confirmation; maintain an open mindset regarding its potential validity until proven otherwise.
15. **Understand the Nature of Dimensions:** Recognize dimensions not just as fixed physical locations but potentially as emergent constructs used to describe degrees of freedom, similar to imaginary numbers in engineering approximations.

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# Full Transcript

So, Jordan, I'm just convinced I'm a string theory ignoramus. You know what? Hearing you say that makes me realize I maybe don't know as much about string theory as I could. But if you know nothing, I can't wait to find out how little I truly know. Coming up, one of the world's experts on that subject. Check it out. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk, Cosmic Queries Edition. I got with me Jordan Klepper. Jordan, Daily Show Jordan Klepper. What would that show be without you? It would still be Daily. They would replace me probably pretty quick. I had no illusions. You know, I'd like to come at you with more confidence. Yeah, cheese meal. What would it be? It would be fine. They would be fine. They get along just fine. Oh my God. They really have a formula. They don't need me whatsoever. Why? Why do you give me such a gotcha at the top of the interview, Neil? Oh, yes. Sorry. Sorry. I take it back. I take it back. And you got a touring show. I just learned. Yes. Jordan Klepper Live. Jordan Klepper Live. Live. And we find you on...

Well, you find me on stage as you'll find my information at officialjordanclubber.com. And then you'll see me up on stages telling stories about my time at the Daily Show, time on the road, time growing up. You'll get a lot of me. All right. All right. That's how we like it. So this Cosmic Queries is all the physics that I never learned in graduate school. So this is sort of an attempt for you to get smarter because you didn't want to pay the extra money or because there was just a failing at the higher education level. Well, so there's a saying that I first heard from my sister who got it when she worked for Michael Dell of Dell computers. Now Dell, whatever, industries, whatever. It's, if you ever find yourself to be the smartest person in the room, change rooms. Terrible advice.

I have worked my entire life to be the smartest person in the room. Here's the opposite advice. If you find yourself the smartest person in the room, stay in that room. Lock the doors. You are king. Enjoy being king. If you find yourself in a room where you are not the smartest person, try another room. There's a lot of rooms. I'm sure there's smaller rooms. There's other

rooms you can find an advantage. Intellect is an advantage that you can wield for your own advancement. get in a room where you can wield that. So potato-potato. So what we have here, we went to Virginia Tech. We found a string theorist, she's been on before, associate professor of physics and an affiliate in the math department. The physics wasn't enough.

Throw in some extra math. It's a definite brag. Oh, that's not enough at a cocktail party. Everyone, help me welcome back to Star Talk, Laura Anderson. Laura, welcome back to Star Talk. Thank you so much. I'm excited to be here. Excellent, excellent. And you not only do research there, but you also teach classes. I do indeed. Yeah. Yeah, there it is. I teach from beginning level all the way up through graduate students. There you go. And it says here, your research includes the geometry and particle phenomenology in string theory. Yes, it does. Duh, obvious. Yeah. I'm mostly interested in the particle phenomenology. The geometry, that's old hat. So we're just going to skip by that. We're phenomenology the entire time. So I got some questions for you.

because occasionally some questions come up that a little beyond my reach or a lot beyond my reach. What's the latest on the Higgs boson? Can you just catch us up on that? Well, the Higgs boson, I would say, is thoroughly discovered in the recent years since the detection of this. This was something that theory had predicted roughly 50 years before. So my experimental colleagues would be annoyed at me for saying this, but by the time of the actual discovery of the Higgs boson, and years ago, roughly 15 years ago, This was something that we really expected to be there. The thing that everybody is waiting for is, is there any physics beyond the Higgs bosong that we can discover with accelerators like the LHC? And that the jury is still out. I'd like your phrase you said thoroughly discovered.

We're feeling really confident on that one. Well, let me interpret that, and you tell me... That's like phraseology. After I lost my virginity, I was using words like that. Mom, I've been thoroughly discovered. No, but thoroughly, I think because when you discover a particle, there's a statistical confidence that you have in that discovery. And how strong is the statistical support for it? So would you say, Lara, that no one is doubting the discovery? Of course, there's been a Nobel Prize given for it. So that would be a good

indication that people are pretty happy with that one. Yeah, excellent. Excellent. And it's properties of the Higgs boson intact. Yes. So as the particle that is giving mass to the rest of the particles in our standard model, Zoo of particle physics, sort of key role of the Higgs boson is very much confirmed in all the experiments that have come since. And yeah, seeing that so cleanly at experiments like the LHC has been a very exciting part of particle physics in the last 20 years. LHC. LHC. The LHC. Large Hadron Collider. Large Hadron Collider. I'm testing Jordan. I know about the Hadron Collider.

Geneva hanging out underground. You didn't let me test Jordan. This is no, please, no acronyms, no acronyms. So when you say physics beyond it, could there be some completely hidden physics awaiting us at even higher energy levels from our particle accelerators, energy levels we have yet to reach? Yep, we certainly hope so. So in terms of energy scale as sort of a resolution of nature, If we think about the wavelengths of light that we see every day around us from things like visible light or microwaves or radio waves, all of those wavelengths are very long comparatively. So the photons involved are lower energy. And if you talk about trying to probe deeper and deeper into subatomic structure, you might have heard of something like x-ray diffraction where you're trying to see inside atoms using x-rays, which are more energetic. That's a distance scale of like 10 to the minus 10 meters.

that x-ray diffraction works. Just to be clear, you can resolve at those scales because the wavelength of x-rays is so tiny that it's commensurate with what you're trying to see so that detail can be revealed. The tool has to match the task. Yeah. Right. And so you need shorter wavelengths. You can't use visible light or microwaves. You got to go down into x-rays. Yeah, so one analogy might be like why can we hear around corners but we can't see around corners, right? And that has to do with the length of sound waves relative to the size of the door, right? Sound waves are bigger than the door so they kind of bend as they go around it and that's why we can hear around doors. Light waves, obviously much smaller wavelengths and they go straight through the door if we don't see around the corner.

Same for when we're trying to resolve this really small structure in atoms, or even smaller, down to quarks or what's beyond that. So the LHC, the big Lertatron Collidery, and this is probing energy scales of TEV, or Terra electron volt, up to 13 TEV. So we can see really small stuff, but not all the way down that we might want to ask. So this is a distance scale of maybe 10 to the 19th meters, but we could try and ask what happens even smaller. Smaller distance scales are higher energies. Okay, so we're we're we're currently blind to that hole to what's beyond what we've already probed That's a the blunt way to say that in terms of experimental direct seeing of it Yeah, of course theoretically we can try and play games and say you know depending on what that structure should be at a smaller scale What would be the signatures of it? How would we know what would happen when you smash these atoms together? That's when the theorists jump in and play. Yeah Now one last question before we go to our a Patreon supporters who each paid 5 to get to be able to ask a question. Well invested money. I think so too. So let me just get up in your face for a minute. Please. Okay. The first murmurs I heard of strength theory was dates back to the 1980s. And that was 40 years ago. 40 years. 40 years.

Albert Einstein comes up with general relativity by himself in 10 years, and we have dozens of string theorists around the globe working on this thing, and you still haven't solved the problem. That is a very, very fair question. Where is this going? Is it leading somewhere interesting? Have we come up with structure that is actually delivering on something worthwhile? You're trying to find the most basic constituents of the universe and account for everything, so it's basically a unified theory of the universe. That's fair, right? Yeah. We're trying to describe all the forces and particles that we see in nature. Okay. And you haven't done it yet. So... This is true. Either that solution is not waiting for us, or it's too hard for our species to figure out, or...

The strength theorists working on it are just too dumb to figure it out. All of you buff. All of you buff. I mean, actually, what you just described there, this question of like, you know, where's this going? Am I going to make a breakthrough? Is it worthwhile? That's something that researchers in any form of research really have to ask, right? You're trying to solve a hard problem and you have to ask, am I making progress? You know, what's this

going to deliver and on what time scale is it worth the trip? And I would say, you know, comparing this to Einstein's theory. Einstein's theory, amazing brilliance, yes, general relativity, one of the greatest intellectual accomplishments of the 20th century. But also in terms of testability, we had a lot of data and measurements ready to go on gravitational physics on the scale of our solar system or nearby solar systems, stars, planets, galaxies that we could try and compare to very rapidly. The hard part about what string theory is trying to do and where I would argue that this progress is sort of innately on a different time scale, is we're trying to describe quantum gravity. And that involves physics from the very, very smallest distance scales all the way up to the large-scale structure and history of the universe. So it's a really hard problem. Is it one that we may not actually solve at all?

I think string theory has already demonstrated that it's a consistent theory of quantum gravity, which by itself is interesting in the same way that we idealize the things like in physics, you know, you pretend that all the shapes or spheres, in order to make the math easier, you pretend that you're in an idealized system where you can ignore air resistance or whatever. You know, that idealization in string theory is very valuable. But the question of whether that theory of quantum gravity exists in our universe, I think, is not one that isn't capable of delivering answers. It's just really hard. It's really hard to push the theory all the way through to decide, you know, is this right or wrong? And obviously I made the judgment call, because this is what I'm spending my time on, that I think we are making progress. I think that there is a hope that this will either be a model of our universe in a way that we can understand, or that we'll learn something about how quantum gravity works, even if it's different than string theory in our universe. Yeah, but no way in there did you say, we're waiting for someone much smarter than everyone else who's working on it to solve the problem the way Einstein did it.

I mean, hey, if there's somebody in the sidelines who wants to come in and solve all the problems, be my guest. We're working on it. All right. I have to get that off my chest. It's a fair question. Thank you for taking us there. So, Jordan. We got some questions. Let's bring it on. All right. And these are people knew you were going to be on this show. And so these questions are, occasionally one leaks in my direction. and I'm happy if it's got some

astrophysics-y things in it, but these came in for you. Well, I'm honored and I'll give it my best shot. If there's any satirical comedy questions, I'm more than happy to take a swig at it as well. I'm counting on you. Yeah, okay, so I'm around. I'm on the bench. You got backup for us. Yeah, yeah, let's see where we go with this. This one's coming in from Germany. This is Josh in Bonn, Germany. In my search for more ways to visualize the madness that is string theory, what actually is a string?

If it's a one-dimensional object, how can it vibrate? It intersects with another dimension, making it not exactly a one-dimensional object. With that in mind, is it fundamentally possible to have any motion and thus transfer of energy without it intersecting at least one higher dimension? Thanks, Neil and Ko. I'm still looking up. That's a good question, Josh. Yeah, the question of dimensionality is something that actually gets like kind of brushed under the rug a lot when people talk about string theory. So we talk about the basic building block of matter, the thing that's replacing these point particle descriptions as this extended 1D object. But that doesn't mean that the space that it moves in is one dimensional. So for example, you know, three-dimensional space or four-dimensional space time, or indeed, as string theory suggests, maybe even more dimensions of space and time, these strings can move through those dimensions. And indeed, the way that we actually describe them isn't even as that one dimension. The way that we really describe them is something two-dimensional called a world sheet.

In relativity, you talk about where something is moving in space as well as time. You might have heard in sci-fi movies, people talk about your timeline. You can imagine a point particle moving forward, making a line through time. If you have an extended one-dimensional string, as that moves through time, it's going to create a surface or a world sheet. In some sense, the strings are actually more two-dimensional in terms of their description and how they vibrate in terms of that space-time description. Even just to confuse matters more, We actually believe that things like the where the endpoints of strings can go are actually part of the dynamical information of the theory So these are things called membranes that can be all sorts of different dimensions and those are also part of how string theory is specified So I'm gonna say my short answer is it's kind of a cheat to say

that string theory is about one-dimensional objects There's actually a lot of different dimensions that go into describing how these things move and how they interact So if you take a string, which is one dimensions, and then move it in a time dimension, and you get a sheet as you were describing, we are three dimensional creatures moving in time. What do we make? This would be some four-dimensional space-time surface for the sort of history of how we have moved through the universe. Okay. Yeah, that's hard to visualize. So you have to bring in the higher dimension. It'd be a fourth spatial dimension.

Three spatial and extra time but the way that we sort of you know plot that is you know you imagine for the particle you would have you know time versus space and there's your line so in order to count the motion in time you need another axis need another direction to talk about and that's what the fourth direction would be is your motion. Okay here's yet another fact from lost in space the latest collaboration between star talk and national geographic. 1995, we discover the first authentic exoplanet, a planet orbiting another living star. So, anyone born from 1995 onward? I now declare each one of you to be Generation X. If you like that fact, you can find 4,999 more in Lost in Space.

5,000 facts to help navigate the universe. Lost in Space is now available wherever you get your books. So Jordan, can you think of higher dimensions in your day? I struggle with even getting past two, so going into this fifth or sixth. No, I mean, I was just told about a book that, what is it? The, not Flat Earth. Flatland? Flatland, yes. Flatland, wonderful book. Yes, which I was like... It should be on everyone's shelf. It should be on my shelf. I probably have it somewhere here, actually. Can you let me borrow it? Promise to return it, Neil. But because I do have such a hard time visualizing this, like the idea of string theory in and of itself is so compelling to me, and yet the idea of getting beyond this fourth dimension is, I don't know what to do visually inside my head with it. So I usually zone out. You win every other human, right? Our brains are not designed to do that. So what do you do? What is your note to a person completely like me? Yeah, how do you do it? What's your trick? Which drug is it?

In terms of being able to visualize more than three dimensions, my brain isn't good at that either. Mathematically, it's not so bad in that the math generalizes really easily to higher dimensions for lots of things. But another way that I like to think about things is in terms of their shadows. So if you think, you know, if I have a three-dimensional object and I look at its shadow on the floor, you know, on a table, right, on a two-dimensional space, you can still learn something from that, you know, 3D thing, from what its shadow looks like in 2D. And so for higher-dimensional things, we can, you know, imagine what would their 3D shadows be like even if we can't visualize their whole four-dimensional shape. Right. So a four-dimensional object casting a shadow into three dimensions would be a full three-dimensional shadow. Correct. Yeah.

that's Plato's string theory cave theory. I think it's the Plato's cave. There's a little bit of that in there. Right, right. So, yeah, this analogy that gets used a lot in these kinds of things is if you imagine the relationship between, you know, a square and a cube, right, you could imagine something that's four-dimensional that has as its shadow something like a cube. And this is where all these cool pictures of things like Tesseracks come in, which are just fun things to visualize in terms of what four dimensions They're three-dimensional shadows, maybe. And they're fundamental to the Marvel Universe, the Tesseract. Yeah, no Marvel powers that I am aware of. But one of the elements of that question was, how can a string accomplish anything if it's not interacting with another string?

In a higher dimension. We expect they do so strings in string theory can they can break apart You can have one string that could break into two pieces. They can break, you know, they can merge into each other Yeah, you can also have things like as I mentioned the endpoints of the strings are actually dynamical So where the strings are that's part of the theory too And so things can kind of dissolve into their endpoints and come back out again And there's all sorts of weird silly putty games you can play with these these And if you take two endpoints and join them, what happens? Depending on the shape of the string, you can have things like closed strings where they form little loops and they can merge together and form bigger loops. You can have a single string that can be open and then become closed. You can join two strings together and get another string of a

different configuration. So lots of possibilities. Okay. However, if you have an open string, you can vibrate it in whatever way you want. If you close the loop, you can't fit an arbitrary set of waves in a closed loop. That's right. It does change the vibrational modes. So just like things like the lengths and configuration of violin strings, as you mentioned, are the standing waves that can exist on them. All of those change for classical strings and they change for quantum mechanical strings. Okay. So they are affected. Okay. Yeah. Just making sure.

Okay. So Jordan, what else you got? I got Ethan Brady here. Ethan Brady says, hey, StarTalk. I'm currently a student at a community college. Recently, my entire class got sparked into a debate due to the following question. If a higher dimensional object intersected our space time, could it appear to us as a black hole? That's such a cool question. I want to hang out in your community science class. That sounds really cool. So a short answer is, yeah, sort of. What we first caveat is we don't know what's happening inside a black hole. Well go find out. Yeah You don't have a free weekend or something to just go sneak out there check it out But take an iPhone with you like take some pics. I've seen the sci-fi films on this Yeah, if anybody is sending you know sending folks. I'm ready But in general relativity, in Einstein's theory of gravity, one of the ways that we characterize black holes is through singularities. So where the sort of smooth trampoline surface of space time that we understand gravity to be, where that starts breaking or getting criss-crosses, creases, spikes in it. And so one thing that is certainly true is that if you have a singularity in some subset of your dimensions. So imagine that you have a singularity that we would see as something like a black hole in three dimensions. That can be, quote, resolved or smoothed out by having higher dimensions available. So one way of sort of visualizing that is if you imagine you had like two lines that cross in a plane, like if, you know, a plane of a piece of paper, where those two lines cross is mathematically a singularity from the point of those curves, because, you know, what happens right at that intersection point doesn't look like a smooth curve anymore.

But if you had an extra dimension where you could pull those two apart, then it doesn't have to be singular anymore. Now you just have two separate, you know, smooth curves sitting there. So the same could be true

for space time and that things that look singular in our universe could actually be smoothed out by having extra dimensions available. So the answer is yes. Yeah, great question and great conversation. But why not take an extra step? Could most of the things we see think of and interact with in our three-dimensional universe, be the shadows of higher dimensions cast into our world? In terms of the size of those dimensions, this gets tricky because we know that if there were big things, big extra directions for things to move in, that we would have seen those with the way that all sorts of experiments run. So for example, in atom-smashing experiments like the Large Hadron Collider, you're smashing two protons together and you watch all these pieces fly out and you calculate their kinetic energy and how they're hitting all these detectors. And if there were large extra dimensions that were the same size as the rest of our universe, things would fly off into them and you'd watch all these pieces just disappear all the time. You'd watch energy and mass disappear.

So we know that doesn't happen. Interesting. So we have bounds on how big those extra dimensions could be based on the fact that we see most of the mass and energy that we experience in our universe contained to our universe. So if the extra dimensions were just sitting there available to the experiment, you would lose information into those dimensions. Yeah, or stuff. You could lose mass, energy, all sorts of stuff. You can lose stuff. I mean, this is, you just, that movie I want to see, we're living in, we are the shadows of another dimension where we're constantly losing stuff. That's where the socks went in the dryer. Where are my keys? Well, how much time do you have? It's a long explanation. Yeah, keys and socks, I think. The dryer is the portal. That's what it is. Right, that's where you lose your stuff. Okay.

So, Jordan, next question. Next question's from William Warren. William says, hello, Dr. Anderson, Dr. Tyson. This is William Warren from Abington, Maryland. If extra spatial dimensions really exist but are curled up too small for us to see, is there any realistic experiment that could reveal them within our lifetimes? Or are they likely to remain forever beyond our ability to test? Thank you so much. Yeah, Laura, why don't you just uncurl them? I mean, come on now. What are you waiting for? So it may not be that they are actually all that small. So we were talking a minute ago about

how we might lose matter and energy into extra dimensions if they were big. But physicists are actually doing experiments in our lifetime, who knows how they'll go, to try and probe what they call large extra dimensions. And the large is kind of funny to non-physicists because we don't mean large like the scale of this room.

We still mean like 10 to the negative 19 meters. But there are actually experiments at things like the LHC where people are doing these atom smashing experiments and then trying to see are there any deficits in energy or outputs from these processes. So is it the case that you could really be seeping some things into extra dimensions? Another use of those extra dimensions that people have considered is if some forces in nature interact with the extra dimensions and some don't. That could explain why gravity is weaker than the other forces, for example. For example, if gravity could seep into those other directions, it's sort of diluted compared to the rest of the forces. So there are lots of potential experiments that could, in our lifetimes, shed light on those scales. In string theory, I'd say the best bet is actually going to be not that, but a more indirect argument, either for or against string theory based on the predictions it makes. I think it's very possible that we could prove that string theory is just straight up wrong.

based on the fact that it just can't get things right about our universe that we already see. So she just slipped something in there that I need some pause. You just said gravity is the weakest force. How weak is it? Very weak compared to the others. How weak is it? How weak? Many orders of magnitude weaker than the other forces. Many orders of magnitude weaker. Yeah. Oh, that's like nothing. Then I got one for you, ready? And I think I verified this arithmetically. If you went to the top of a rocket that we launched into space and removed all the electrons from the top one cubic centimeter and put them at the bottom of the launch pad, that leaves a net positive charge in that cubic centimeter. And you have negative charges down there. What did negatives and positive charges want to do? What do they want to do?

What do they want to do? They just want to hang out. Together. They want to be together. Together. They love to bond. Okay. Their force of attraction is large enough to prevent the rocket from launching. It's amazing. That is

amazing. Yes. Yes. So gravity is really freaking weak. You guys are dogging, Graven. I mean, come on, gravity's worked. I'm not really dogging on it. Gravity's so weak, I can reach down. How weak is it? And I can just pick this up. I have the entire earth pulling on this, and I can just pick it up, just by bending over. It's actually a really good thing for all the stuff in our universe, right? Because you couldn't have things like tables and chairs and people that are all held up by electrical interactions, if that wasn't true, right? If gravity was stronger, we'd all just get pancaked. Yeah, the table wouldn't hold. The table's held together by electromagnetic force, your held together by electromagnetic force, the cup is held together by, and gravity does not collapse it into a pile of goo.

Gravity is much stronger than you guys give it credit for. No, it's not. In high school, I wanted to dunk a basketball and I will tell you I tried my hardest over and over again and gravity thwarted me every single day of high school. Is that right? How big are your hands? I have big hands. It has nothing to do with my physique. It's gravity. Those are big hands. Okay, are you able to, could you ever dunk a basketball? When I was in ninth grade, I could palm a basketball and dunk the ball. Why couldn't you now? I didn't say I couldn't now, but I said when I first did it. I can't now. Because of gravity. Okay. So I'm curious about this, that you can lose strength leaking into other dimensions. And I'd heard, and I don't know why, because I didn't take these graduate physics classes, I heard that electromagnetic forces are contained within our universe.

But gravitational forces can leak out and be felt by an adjacent parallel universe. Is that true? Or at least extra dimensions, yes. So the sort of theoretical framework we're describing isn't necessarily string theory, it could exist outside of string theory, and that's a proposal for maybe why gravity is weaker, why the hierarchy problem is it's called in particle physics, you know, that separation of scales between gravity and the other forces exists. Could what we call dark matter The ordinary matter in an adjacent universe, but its gravity is spilling out into our universe, and we're saying, oh, we have a magical force here, it's dark matter, but it's just regular matter spilling into our universe. It's a fair question, and we don't know what dark matter is, so, you know.

all all bets are on the table at this point. I think there's a lot of proposals for this but no one is sort of universally agreed upon in terms of you know what would be the definitive answer like yes for sure this works. So if you can see one of these is these large extra dimensions in an experiment that would be for sure a great confirmation. The other way you can try and do it is to ask a question like you just did and say, is there some signature either in terms of particles or forces or interactions that we would see if that explanation is right, gravity is weak because it's leaking into another dimension? What else would we see in an experiment? What would be the consequences of that prediction? And if you could make concrete predictions that you can then go test, then you're up and running. All right, I'll get right on that, okay? And yeah, spoiler alert hasn't happened yet, so this is still an open question. All right, give me some more. Here's what we got. We got it from Jay Valliano. Dear Dr. Anderson, is it possible to explain the consistency test, d minus 10 equals 0 in layman's terms? And how come some theories speak of 11 dimensions if the formula is 10? Thank you in kind regards from Holland.

So, first off, nobody when they were first trying to understand string theory wanted extra dimensions. They would have been very happy if we could just keep our three spatial and one time-like dimension. But the origin, let's start with the D equals 10 in string theory. So string theory is this idea that you can, you know, you can put it on a t-shirt. It's very concise to say that all matter energy in the universe of these little vibrating strings, right? Sounds really great. But then you try and say, quantum mechanically, What would that mean? Like if I have these little quantum mechanical strings, how would they move? How would they behave? Could they be consistent? And it turns out that theories, before you look at the sort of smallest quantum mechanical distance scales, they make promises. So some of those promises might be things like electric charge should be conserved, should not be created or destroyed in interactions.

Quantum mechanically now you have this probabilistic theory where all sorts of you know uncertainties and possibilities and sort of quantum weirdness takes over and those classical promises can get broken so you could have you know the conservation of electric charge is being violated quantum mechanically and it turns out that when that happens it's called an

anomaly in quantum field theory these anomalies goof up the bookkeeping of your theory so. It's like a bank where you're having pluses and minuses, adding up all the tallies. And if you don't keep the math right, if all the numbers don't add up, then it's not just that the theory is going to make a bad prediction. It's that the theory just doesn't even make sense. It's not consistent. It's not a useful tool for the things that you wanted it to do. So when people started studying these string theories 40 years ago, they tried to consider these quantum anomalies and asked whether they were really fatal, whether they were going to kill this whole line of research. And what they found is that All of this quantum uncertainty, this sort of wiggling and violation of these principles that we think should hold, magically cancels out when you have 10 dimensions. These quantum anomalies cancel exactly. So the theory becomes really robust and really rigid. It starts making all these sort of theoretical structure and predictions at that point. And then you can ask questions. You can say, well, what does this predict? What does it mean? Can it work? And we're still trying to figure that out.

But at least you have a mathematically internally consistent theory at that point. So that's where the D equals 10 comes from. I love the phrase quantum anomaly. It just feels cool. It feels like a Kendrick Lamar line. It's just playful. It's fun. It's specific. Yeah. Yeah. Yeah. All of the above. I want to stay on this side of the the earth. Near Holland. This is from northern Sweden. OK. I think that's I think that's near Holland. You're American, so you have no clue. I'm going to say it's close. This is from Frederick Johansson. Why is it so difficult to access the higher dimensions? Do you think we'll ever get to see or manipulate them in any way? People really want to know. Yeah. People want to see. Are we actually going to see this effect? Are we going to be able to prove it in our lifetimes? What do you say to Frederick? Let me give another lead into that. We're familiar with the story, the book Flatland by Edwin Abbott.

I think that was published back at the turn of the previous century around there. And it chronicled the encounter between people who lived in a flat universe, these squares and triangles and other shapes. And then they encounter a three-dimensional shape, and they freak out and don't know how to explain it. And there's a hierarchy of how many sides you have as a measure of how aristocratic you are. Of course, I have to throw that in there

as well. My question is, if you're a square in Flatland, what's preventing you from finding that direction perpendicular to your flat universe to just get a peek at three dimensions? What's stopping us in three dimensions from finding that portal to get a peek at fourth dimension? Why are you preventing us from doing that? That's a really good question.

You know, you're exactly right. So in the Flatland analogy, if you just had two dimensions and there's a third just sitting there, yeah, your little 2D, you know, flat people could just pop out of that universe up into a three-dimensional space. And indeed, if you let things go, right, you just let them, you know, move, have kinetic energy, bounce into each other, have collisions, if you have large extra dimensions, you would do that. You would see them, you would move into them, you know, you can't prevent or, you know, force things into a restricted set of dimensions without a lot of effort. The catch with the extra dimensions as they're arising in string theory is this notion of scale again. So imagine that we are three-dimensional beings. Imagine that you're walking on a rope, right? And you can walk down the length of a rope, tight rope balancing. If I say, can you walk around the radial direction or the circular direction of the rope?

you know, rope compared to our feet, right, is really hard. We can't really resolve in terms of our motion that extra two-dimensional surface of the rope. If you imagine zooming that even further, you know, if you look from very, very far away, that rope looks essentially one-dimensional. So the problem is just all the interactions that we're measuring are on such a big scale that they couldn't access these extra directions that we're talking about that would be wrapped up around our existing universe. Okay, so she... She weaseled out of that one, too. Did you hear that? You hear that weaseling? You're saying we can't get to those other dimensions because they're so tiny. But if they were big, are you then allowing us to say that they would come into view and we can step into them? Yeah. And as I mentioned before, like.

People, they actually think that maybe these extra dimensions are big enough that we could see them in particle physics experiments. So that's all it has to do with. How big the dimension is. Do we have certainty on the size of the dimensions? We know that they can't be bigger than certain energy scales. So the experiments at the LHC, for example, are probing

around up to a few TEV of energy or roughly about 10 to the negative 19 meters. So we know that they're not bigger than that or we would have seen them already. So a TEV, just to catch us up here. So an EV is not an electric vehicle. It's not. No, not in this kind. It's an electron volt, which is a unit of energy. And it's the energy... I haven't done this since like physics 102, but it's... I like that even as you're being humble about it. It's not physics 101. It's physics 102. Just to be clear, I skipped 101. So...

is the energy it takes to move an electron, which has a charge, across a potential difference of one volt. That's an electron volt. Did I get that right? I think so. Oh, right. Yeah, I remembered it. OK. So that's helpful when you're talking about particles, because it's native to their world, right? And so a teraelectron volt is 10 to the 12 of these things. So that's a big number. That's a big number. And that's the energetics of our particle accelerators today. Yeah. I get thinking about these different wavelengths of light and what you can see just using photons as an example. You know, you think about radio waves, right? Like how big does a radio tower have to be in order to make radio waves? Like it has to be the size of a building roughly like tens of meters. That's, you know, the length scale that you're probing if you were to bounce.

radio waves off things. And the grading on your microwave has to be millimeter-sized to keep all the microwaves inside your microwave box. And now we're talking about distant scales of less than 10 to the negative 19. Those are really energetic things, photons, particles, whatever that you need to be able to probe that low. So it's hard. There are ways, as I mentioned, though, indirectly to try and either prove or just prove the theory. And lots of clever theorists are trying to conjecture or come up with sort of no-go results that could proof-string theory wrong in other ways. So could we see extra dimensions in our lifetime? Maybe. And people are looking. That would be great if we do. And even if we don't, we might be able to decide some of the merits of this in other ways. I just like the idea you can make a dimension big enough to Jen to say, I want to visit that. I mean, you'd want to. You'd want to. And it's got to be completely mind-blowing. Like a square looking at a cube for the first time, that's got to be mind-blowing. Yes. And how disappointing from the other direction.

Is that all I am a square? Yeah, a cube looking at a square is like, oh, okay. Well, you do that all the time, looking at your shadow. You're right. I'm always, and I'm always disappointed. There are great animations online, by the way, of three-dimensional shadows of four-dimensional stuff. Definitely worth checking out. Oh, I've got to find them. Oh. All right, all right. Give me a couple more here. Okay, Trip Knee Salo says, uh, hi there, Neil the Gang. This is Salo from the Philippines. Uh, string theory says there are hidden extra dimensions curled up all around us. But, uh, holography? Holography? That's right. There's the holographic universe? Is that the... That's... Yeah, well, but holography? It's not holography. Holography? Yeah. Yeah.

We'll fix it and edit. Edit around this. Make me look smart. Fix it and post. All right, yes. It suggests our entire universe is just a projection from a flat 2D boundary. Is it possible that both of these theories are actually right? Are those extra dimensions just part of that projection? More power to the show, and I hope you can visit your Filipino fans soon. Oh, very nice. I've actually never been to the Philippines, but I'll put that on the list. You got a friend there? Yeah, yeah. So I don't know that we are sure about whether we're living in a holographic universe or on the surface of a black hole, but the underlying physical idea that I think is really exciting is that things like gravity or things like gauge theories, which describe things like electromagnetism, seem to be very deeply related in their structure in ways that no one had anticipated many decades ago.

One utility in a physical theory is that if you can simplify things, if you can come up with a really simple framework that could make a lot of cool predictions, that's useful. So one of the things that's really neat about the holographic principle is if you take, for example, our three-dimensional universe or four-dimensional space time, if you said that all the dynamics or things that are moving in our core universe, if that could be simplified to something that's happening on the boundary of our universe, if we could describe it that way, that'd be really powerful. And there are in fact people that are trying to answer exactly that question independent from string theory. There's a whole program known as celestial holography, which sounds really cool. I'm talking about the celestial sphere at infinity in our universe and how all the information of things like gravitation in our

universe might be encoded on that boundary. So it's a theoretical tool, but it might be a very useful one long term.

Yeah, but how you saying I'm alive on the boundary of my own universe? Or I guess the angle I was pushing it on is we can get information about everything in the universe from its boundary and that might be simpler. It might show us a sort of hidden core of things or an organizing principle that might help us solve problems we couldn't have solved otherwise. That's wacky stuff. I was gonna say it sounds philosophical and poetic that it's on the boundaries are the only ways in which to see sort of what we are existing within. That does sound good. Also a Kendrick Lamar line. It's very Zen Zen of you. It is maybe for Alan Watts. Yeah. This is from Farouk. Greetings. This is Farouk from Queens, New York. The question has been on my mind for the past couple of years regarding the nature of time.

If time is measured by the displacement of atoms, but those atoms stop moving, does time still exist as a fundamental property? Can I take a stab at this? And you clean up my mess? So I think there are two ways you can reckon time. Maybe many, but two that aren't I think of when I think about that question. One of them is in your own sort of world, you can sequence events. Now this is not for every observer, but for you you can sequence events to know what happened before or what Okay, so that's in a sense of time you have a before and an after but I can tell you this if you have no thing that repeats itself Then you have no way to measure the passage of time Wait explain that to me earth rotates once a day and it keeps doing it earth goes around the Sun once a year and it keeps doing it We need the we have a day in a year Okay? We got a clock. We got atoms that vibrate. Everything is repeating. The measurement of time is embedded in repeated phenomenon. So if you go to a world where nothing repeats, there would be no measurement of time. Doesn't mean there isn't time still existing, but you couldn't measure it in any meaningful way. Are we good with that, Lara? Yeah.

No, I think you're saying that you agree with the questioner, which I think I would too, yeah. In a totally static universe, that isn't a really meaningful question, right? If nothing's moving, if nothing's evolving, then, yeah, you don't really talk about time. Well, okay, so let's get, take it another layer in.

If nothing's moving, everything is static. All I said was that you can't measure time. But you said time won't matter, which is surely true, because everything is the same at all times. If nothing changes, what do you care what time it is? But you still experience time. That's what I'm asking. So, is there still a reckoning of time if there's nothing to measure it? I mean, I think if nothing is changing literally, then even the measurement or perception of time doesn't mean much, right? One thing that we do know for our universe about time is that time, as you were alluding to in your answer, you were saying, you know, an individual person experiences time we have a before and after.

But there is no universal clock for our universe. We know that in special relativity, all across the universe, in gravitational fields, in different parts of the universe, people are going to measure time differently. Time is going to tick differently. So then you can say, yeah, like, what does that mean if this is a thing that's changing everywhere? What do we mean by this sort of fundamental notion of time? And there, what we know is that it's not just time, but it's particular combinations of space and time. that the entire universe has to agree on. So it may not be pure time, but there is something that everybody has to agree on for universes that do have things moving. And that's given to us by special relativity. I love that. I do. Time ticks differently. What? I know. And if nothing changes, you really have to ask, why do you give a shit about time at all? I mean, think about it. It's like maybe work on changing some shit.

Oh, you're worried about time? I don't know. Why don't you clean up your mess? It's been like this forever. And then you break the symmetry of the universe, and then time begins. Now you got your time, are you happy? Because you cleaned your room. This is from David de Leon. Greetings and salutations, Dr. Tyson and Dr. Anderson. David de Leon. Oh, he gives us a little. Oh wow, we have a lot of info. David DeLeon raised in Florida, writing out of Indiana. It may just be my own ignorance, but we as humans talk about dimensions in multiple ways. More often than not, I assume it is the physical. The physical three, length, width, height, essentially representing up, down, left, right, forward, and back. Would it not be the most logical that the fourth dimension would be out and in or quantum and cosmic? I often like imagining zooming out or in at unprecedented

distances and winding up in a scene similar to that where the zooming started like a universe in quarks. Any thoughts? Again, thank you for all that you do. Big fan of all knowledge and interpretations of life. That's a good question. I would say it's a little different, but the sort of zooming out picture that the question is alluding to is a real thing in physics and it matters a lot.

And this has to do with several things. So the first is whether you expect all of the physics, let's be broad about it, to be the same at every distance scale or not. So if you were to say, I measure something here in my little part of the universe to be this big, and now if we just scaled it up arbitrarily smaller big, should all the laws of physics be the same? And we think that that isn't actually a symmetry of the laws of physics, as we understand it. We think that the structure is different at different energy scales. That symmetry, if you had a scale where, like, length didn't matter like that, you could zoom in and out, up and down, and everything was the same. That's called a conformal symmetry in theoretical physics.

And some things do use it in a really powerful way. So actually, these string world sheets that we mentioned before, these little one-dimensional strings that are vibrating through space, those have a conformal symmetry. You should be able to scale that world sheet up and down, and it doesn't change how that world sheet behaves, which is really neat. But other physics, like the physics of the strong interactions and quarks, really does seem to be behaving on a different energy scale than gravitation. And this gets studied in things like quantum field theory a lot, where you try and say, If I'm writing down a physical theory, what regime of scales and sizes and energies should it be valid for? So it's just to be clear, it's not that the laws of physics are different, it's that they manifest in different ways. Or the way that you describe them, the systems of equations and the descriptions that you have do change with those different scales. Okay, so we're not just zoomable. Yeah, although as we mentioned, string world sheets are, which is kind of cool. So some stuff is, some stuff isn't. We still like to be.

It's still a fun idea. Thank you for putting it out there. Dear Dr. Tyson and Laura, this is Jonas William from NYC just down the road. My question for you is, since string theory dualities are so hard to calculate in smooth space,

could we use the sine vectors of oriented metroid to build a pixelated discrete model of spacetime? Would stripping away the smooth geometry naturally solve the infinite singularities of quantum gravity? That's a great question, and lots of people are actually thinking about that. So people have hoped that there was some sort of fundamental little unit or building block, like building up a big tiled floor by having lots of little tiles that you fit together. Maybe you could tile your way to a complete picture of space time. And there are several different types of physics that are exploring that. So things like loop quantum gravity or something called causal dynamical triangulations, try and do exactly that. They try and build quantum gravity through discretizing space time. And I'm not an expert on either one of these fields. These are not string theory, but there are different things that people are exploring. And I think it's a perfectly fair question. Does that solve the problem of quantum gravity? I would say the jury's still out. We haven't decided, but more power to people for asking the idea. And who knows, maybe Jonas, you could figure that out too.

Even Neil's like, I don't know. Let's see if we have one final big question that we can sink our teeth into here, all right? Keith from New York City, Dr. Anderson, is it necessarily true that dimensions are something that objectively exist in nature, or are dimensions simply an organizing principle that we puny humans use to make sense of the universe? Are they features of reality, or could they be emergent, useful descriptions that arise from something deeper? Is it even possible that dimensions are not real in any sense, but are instead conceptual tools, much as we use imaginary constructs, orbital model of the atom, imaginary numbers in physics and math? Thanks. That's a great question. I would say, yeah, I mean, most of the way that we describe the universe is a construct. So could dimensions just be a construct? Yeah.

Certainly things move, and we can see the degrees of freedom of how things move in our universe. And so that's where you see that you can move forward, backwards, side to side, up and down, and those are different. But labeling them as dimensions, labeling them as different, that's just a tool for how we try and describe physics and make predictions. Just like things like a force when you say something pushes on something else. That's a construct too. Turns out a very useful one. But we don't measure forces, we

just measure acceleration. Yeah, I would say it could be viewed as a construct, but one that we think we can get a lot of mileage from. Because that reminds me in some branches of quantum physics that we don't understand, we just like the hidden variables hypothesis. It's like they just assert it because it allows you to think classically about the problem, whether or not it's actually true.

But there have been occasions in the past where people just pulled something out of their ass to just kind of account for it and it turned out to be true. There are a couple of occasions such as that. And some things where, no matter how many hidden variables you have, you still can't get it right, like in quantum mechanics. There's theorems that say no matter how many extra degrees of freedom you have, could you erase that quantum weirdness, that uncertainty, the probabilistic nature of quantum mechanics? And the answer is no. you could have an infinite number of hidden variables, degrees of freedom, doesn't help. Fundamental to it. Well. I think we learned some stuff here today. These were great questions. Your audience is really cool. They're my people. We got good people here. And global. This is an international crew here. People want to know, you know?

Inquiring minds want to know. Absolutely. Inquiring Patreon members who want to know. So, where can we keep in touch with your work? Do you have a website or...? I have a website? I am not on social media, which is a bit embarrassing. I'm a little bit of a let-it. That is a dimension you don't need to explore. I think you have enough on your plate. I'm a little bit occupied otherwise right now. Yeah. Okay, so not on social media, but you have a website. Which is it? Averginitech. You just search my name and physics. You'll probably find me. All right. Well, Laura, thank you. Thank you for sharing your cosmic expertise with us, living completely in the gaps of my knowledge. Well, thank you so much for having me. This was a great pleasure. And yeah, I think your audience members are awesome. So I'm delighted to have a chance to chat. And we think so too. All right, dude. Neil, thank you. Have you always? Always good. This has been another episode of Star Talk. I'm Neil deGrasse Tyson, as always bidding you.

to keep working up.